

ORIGINAL ARTICLE OPEN ACCESS

Comparing Watershed-Based eDNA Sampling and Camera Trapping for Assessing Mammal Diversity in North-Western Bhutan

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Received: 9 July 2025 | **Revised:** 28 November 2025 | **Accepted:** 6 January 2026

Keywords: Bhutan | camera trapping | eDNA | mammal | metabarcoding | watershed

ABSTRACT

The ability of environmental DNA (eDNA) to provide rapid assessments of mammal taxa composition at the watershed scale can make it an efficient survey method on large-scale landscapes, complementing camera traps. Due to the rugged and inaccessible terrain of many areas in Bhutan, camera trapping is associated with logistical challenges, increasing the cost of sampling considerably. In this study, conducted in the Upper Punatsangchhu catchment basin of Bhutan, we investigated the ability of eDNA water samples to capture the diversity of terrestrial mammals in comparison with camera trapping, using six watersheds within the basin as a baseline sampling frame. Combined, the two methods detected a total of 72 mammalian species: eDNA metabarcoding identified 60 species, while camera trapping detected 33 species, with an overlap of 21 species between the two methods. In addition, eDNA metabarcoding detected 90% of the IUCN Red List species detected by the camera traps. Small mammals were frequently detected using eDNA metabarcoding, while camera trapping more often detected large mammals. The mean detection probabilities recorded from eDNA were higher for all species grouped by orders and size categories compared with camera trapping. Biodiversity models based on eDNA metabarcoding and camera trapping both retrieved dominant effects of temperature and isolation in structuring the mammal assemblage. We conclude that eDNA sampling based on watersheds accurately represents the spatial distribution of species across each watershed in our study area in Bhutan to provide a rapid assessment of mammals from river water.

1 | Introduction

Terrestrial biodiversity is fundamental to ecosystem stability and the provision of ecosystem services, such as pollination, water purification and climate regulation (Blackman et al. 2022; Seren and Celekli 2024; Tilman et al. 2017). A number of anthropogenic factors are threatening this biodiversity,

increasing the likelihood of species extinction on a regional and even global scale (Bellard et al. 2022). Habitat fragmentation brought on by land-use changes, such as urbanization, agricultural expansion, and infrastructure development, is degrading the quality of habitats, especially for wildlife. Moreover, pollution and overhunting directly reduce species populations (Tilman et al. 2017). Global changes are leading

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to an alarming global species extinction, as well as local extinctions of numerous species (Altermatt et al. 2023; Seeber and Epp 2022; Seren and Celekli 2024; Shivanna 2022). To halt wildlife erosion, various conservation actions have been implemented, including the establishment of protected areas and the restoration of degraded habitats. Effective conservation requires efficient monitoring of wildlife to assess the impact of management strategies. However, monitoring wildlife species using traditional approaches is generally time-consuming and costly, so quantifying the effectiveness of these measures is difficult and prevents their optimization.

Addressing the challenges of biodiversity loss requires regular monitoring of species and ecosystems, which enables early detection of declines or ecosystem imbalances and evaluations of the efficiency of measures targeting wildlife protection. By using biodiversity monitoring data, scientists can assess conservation efforts, adapt strategies, and allocate resources effectively, ultimately strengthening biodiversity resilience (Kennedy et al. 2023). In this context, monitoring terrestrial mammals is particularly important, as they are among the largest extant animals and often dominate various ecosystem-level processes, such as energy flow, nutrient cycling, and vegetation dynamics, due to their high energy demands (Lacher et al. 2019). Camera trapping is widely used for surveying terrestrial mammals and monitoring their populations, movements, and distributions (Leempoel et al. 2020; Mills et al. 2019; Shermeister et al. 2024). However, this approach is associated with several technical and logistical challenges. In large-scale surveys, setting up cameras, checking batteries, replacing memory cards, and retrieving equipment are time-consuming tasks that require extensive planning and can be costly (Barros et al. 2024; Lyet et al. 2021; Sales et al. 2020; Seeber and Epp 2022). Further, the use of camera traps is often limited by problems such as damage caused by wildlife, theft, and vandalism of the equipment (Barros et al. 2024; Glover-Kapfer et al. 2019). In large-scale sampling, camera trapping often fails to cover certain areas, and conservation decisions are usually based on fragmented data that do not fully represent entire habitats (Lyet et al. 2021). The difficulty of using camera traps across wide areas and the need for rapid time-series data emphasize the importance of finding alternative wildlife monitoring methods.

The use of environmental DNA (eDNA) metabarcoding has revolutionized biodiversity monitoring, complementing conventional methods (Sales et al. 2020). eDNA corresponds to DNA released by organisms in the environment (Taberlet et al. 2012). Sampling for eDNA involves collecting genetic material shed by organisms in the environment, such as skin cells, hair, feces, saliva, urine, or bodily fluids from various substrates, such as water, soil, or air (Harper et al. 2019; Leempoel et al. 2020). Sampling eDNA from water for the detection of mammalian species is opening new possibilities for studying their distribution over extensive spatial and temporal scales. Moreover, networks of streams act as natural pathways that carry DNA across different landscapes (Lyet et al. 2021; Mas-Carrió et al. 2022). eDNA metabarcoding enables the measurement of biodiversity and species presences in river water across entire catchments, offering a cost-effective and time-saving approach (Lyet et al. 2021). Additionally, eDNA metabarcoding facilitates monitoring in

places that are difficult to assess, such as dense forests or rugged landscapes, where conducting physical surveys may be impossible (Mena et al. 2021). On a large scale, however, determining the origin of eDNA signals to accurately represent habitat use by different species remains challenging (Urycki et al. 2024). Therefore, to determine whether detections accurately correspond to the catchments or watersheds being tested, an appropriate sampling design is necessary.

Bhutan is known for its rich biodiversity, which is linked to the nation's conservation philosophy. Located within the Eastern Himalayan biodiversity hotspot, Bhutan is home to a high number of threatened and endemic species (Devkota et al. 2023). The country has committed to preserving its natural wealth through policies that prioritize environmental conservation, including a constitutional mandate to maintain at least 60% of its land area under forest cover (Bruggeman et al. 2016; Lham et al. 2019), directly protecting the habitats of numerous wildlife species. Challenges persist in safeguarding important mammalian species due to habitat degradation, human–wildlife conflict (HWC), and poaching (Wangchuk and Dendhup 2024). Monitoring wildlife effectively is crucial to addressing these threats. Until now, camera trapping has been the main method for monitoring mammals in Bhutan (Lham et al. 2019). Conducting this type of sampling at regular intervals to establish a time series of species detection has been difficult, due to the lack of consistent funding, which poses challenges for the continuous monitoring of species (JDNP 2021). To date, data from camera trapping have remained the primary source for informing policies and conservation actions, particularly in habitat protection and community-based conservation efforts. Considering the shortfalls of relying solely on camera trapping at the country scale, eDNA-based surveys are seen as an important approach to complement wildlife monitoring in Bhutan and further strengthen conservation activities.

In this study, we evaluated the capability of watershed-based eDNA sampling from rivers and streams to detect mammalian species, in comparison to camera trapping. We selected a watershed in north-western Bhutan, which includes the entire Jigme Dorji National Park (JDNP) in the north and part of the transitional zone outside the park in the south. JDNP has rugged terrain, and most parts of the park are inaccessible (Dendup, Wangdi, et al. 2021). Access presents major challenges for species monitoring, particularly due to logistical constraints and high costs. Despite these challenges, the Department of Forest and Park Services (DoFPS) of Bhutan has conducted extensive camera trapping in the past (DoFPS 2023; NCD 2023), resulting in a comprehensive species presence data set that can be used for comparisons with eDNA detections. In this research, we aimed to answer the following questions: (1) How does the detected taxa composition differ between eDNA and camera trapping? (2) Are there differences in detection probabilities between taxa for eDNA and camera trapping in relation to species body size? (3) How does eDNA-based taxa composition vary between sampling sites and watersheds when compared with data from camera trapping, and do the identified environmental drivers of taxa composition differ between methods? Overall, we aimed to determine future sampling strategies for eDNA-based surveys, using watersheds as ecological monitoring units.

2 | Materials and Methods

2.1 | Study Area

We conducted the study in the Upper Punatsangchhu catchment basin, which encompasses the Jigme Dorji National Park (JDNP) in the northern region and an area just outside the park in the southern parts (Figure 1). Spanning an area of 4374.06 km², JDNP is the second-largest national park in Bhutan, stretching spatially from 27°33' N to 28°15' N and from 89°14' E to 90°22' E (Dendup et al. 2023). The park's elevation ranges from 1200 to 7200m (Thinley et al. 2016), while the adjacent southern area has an elevation range of approximately 1200 to 3500m. Due to its wide elevation range, the study area covers diverse vegetation types. Above 4000m, alpine habitats dominate, while sub-alpine, mixed-conifer and broadleaved forests are prevalent below this elevation (Thinley et al. 2016). The region serves as a critical habitat for several threatened mammal species, including the dhole (*Cuon alpinus*), musk deer (*Moschus chrysogaster*), snow leopard (*Panthera uncia*), and royal Bengal tiger (*Panthera tigris tigris*) (Dendup, Dorji, et al. 2021; Dendup et al. 2023; Leki et al. 2018). JDNP is the only known habitat worldwide where both apex predators, the tiger and the snow leopard, coexist within the same habitat (Dendup et al. 2018). Human communities within the region

display varied livelihoods influenced by elevation. Inhabitants above 4000m follow a semi-nomadic pastoralist lifestyle, with a significant focus on yak husbandry (Dendup, Dorji, et al. 2021; Leki et al. 2018), while lowland communities below 3000m are engaged in mixed farming practices, with agriculture taking precedence. Despite its cultural and ecological significance, the area faces numerous conservation challenges due to anthropogenic pressures and environmental threats (Dendup, Dorji, et al. 2021).

2.2 | eDNA Sampling Design

The Watershed Management Division (WMD) of Bhutan has delineated 186 watersheds within 5 major and 2 minor river basins across the country to assess their status and condition (WMD 2011). In this study, we selected six watersheds—Lingzhi, Laya, Goenshari, Phochu, Thimphu, and Dangchu—and established 1 to 7 eDNA sampling points in each. These points were strategically located to cover the mouths of the main river in each watershed, as well as the accessible upstream tributaries (Figure 1). This watershed-based sampling design ensured that all parts of the overall study area were substantially represented. Overall, the eDNA sampling sites covered the entire Upper Punatsangchhu catchment basin. Between July and December

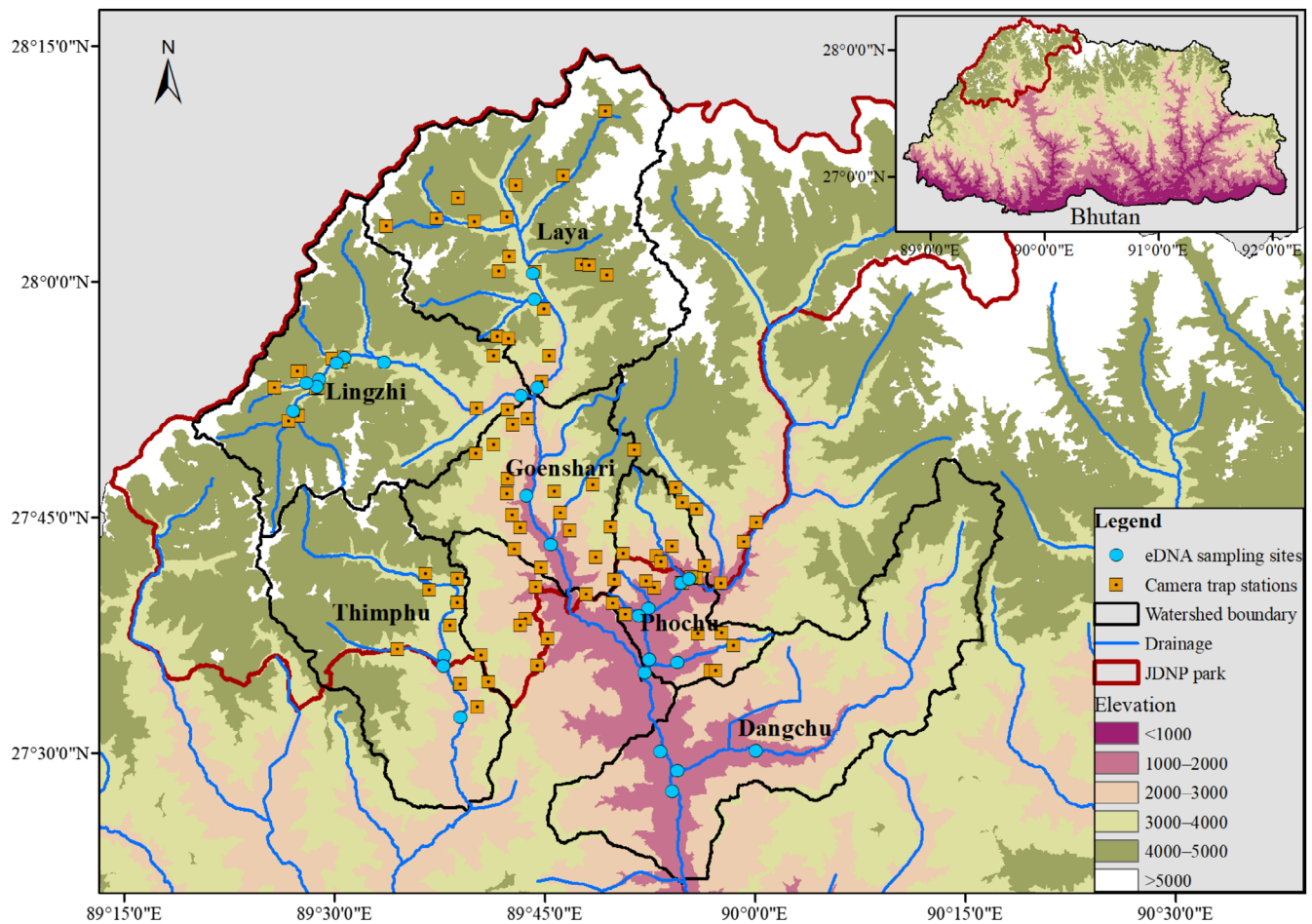


FIGURE 1 | Map showing the environmental DNA (eDNA) sampling sites and camera trap stations. The camera trap stations include those acquired from the Department of Forest and Park Services (DoFPS), as well as those deployed for this study. The eDNA sampling was extended beyond the Jigme Dorji National Park (JDNP) area to include a mosaic of lowland habitats. Moreover, these areas serve as an important ecological corridor to the nearby protected areas, facilitating the movement of species between them.

2023, we collected a total of 66 eDNA samples from 27 sites, with 2 to 5 replicates at each site. The eDNA sampling involved in situ filtration of 30 to 60 L of water, using a portable peristaltic pump to filter water through a large-capacity Darlly filter fitted with disposable sterile tubing (Chacko et al. 2023; Lyet et al. 2021; Mas-Carrió et al. 2022), running for 1 to 2 h. The filtration lasted less than 30 min for 19 filters due to sediment clogging. The volume of water filtered was measured with a 1 L water bottle at the interval of every 10 min. During the filtration process, we minimized the risk of contamination by changing sterilized gloves at each step of the filtration process. We collected samples from the fastest-flowing portions of the rivers or streams accessible from the riverbank (Lyet et al. 2021). After filtration, we filled each filter with 50 mL of Longmire buffer solution (ELE, ETH Zürich, Switzerland) to preserve the DNA content of the sample. Each filter was sealed in a sterile, individually labeled plastic bag to prevent cross-contamination between samples. Samples were stored at room temperature until further laboratory processing.

2.3 | Camera Trap Sampling

To compare detections from camera trapping and eDNA sampling, we used both historical data and a new camera trap data collection campaign. Past camera trapping data were part of nationwide tiger and snow leopard surveys (DoFPS 2023; NCD 2023) collected by the Department of Forest and Park Services (DoFPS). The DoFPS tiger survey focused on elevations below 4500 m, deploying cameras within 5×5 km grid cells for the duration of 4–5 months (DoFPS 2023). For the snow leopard survey, camera traps were placed at elevations between 3400 and 5200 m within 2×2 km grid cells for the duration of 2 months (NCD 2023). To complement the camera trap surveys and strengthen the comparison with the eDNA data, we deployed a total of 31 camera traps between July and December 2023 for the duration of 6 months: 14 in the Lingzhi watershed (due to low detections from past camera trappings) and 17 in the Phochu watershed (due to a lack of data). We deployed each camera trap in an accessible area within a radius of 5 km from an eDNA sampling site, maintaining a minimum of 2 km between the camera traps. Site selection for these camera trap stations prioritized areas with high potential for species use, such as fresh trails, rock crevices and water holes. Due to time limitations, we chose not to install cameras in the Dangchu watershed, located outside the JDNP. We combined observations from the past camera trapping data and the new deployments into a presence–absence matrix to compare detections between camera trapping and eDNA-based sampling. We carried out camera trap file management using Camera Trap File Manager v2.1.6 (Dou et al. 2019), manually identifying each species.

2.4 | eDNA Extraction, Amplification and Sequencing

We performed the eDNA extractions in a dedicated sterile laboratory equipped with negative air pressure to prevent any sample contamination during the extraction process. We followed the extraction protocol outlined by Pont et al. (2018) and Polanco Fernández et al. (2021), with modifications. Specifically, we

agitated each filtration capsule, containing 50 mL of Longmire buffer, for 15 min on a S50 shaker at 800 rpm. We then transferred 30 mL of buffer to 50 mL tubes and centrifuged them at 5500g for 15 min. We removed the supernatants with a sterile pipette, leaving 9 mL of liquid at the bottom of the tube. Subsequently, we added 19.8 mL of absolute ethanol and 0.9 mL of 3 M sodium acetate to each 50 mL tube, and we stored the tubes for at least one night at –20°C. We then centrifuged the tubes at 5500g for 35 min at 6°C and discarded the supernatants. Next, we added 720 µL of ATL buffer from the DNeasy Blood & Tissue Extraction Kit (Qiagen, Hilden, Germany). We vortexed each tube for 5 min and then transferred the mixtures to 2 mL tubes containing 20 µL of proteinase K. We incubated the tubes at 56°C for 2 h. Subsequently, we performed DNA extraction using the NucleoSpin Soil mini kit (MACHEREY-NAGEL, Oensingen, Switzerland) starting from step 6, following the manufacturer's instructions. We performed the elution by adding 50 µL of SE buffer twice. DNA extractions were performed in five batches, each containing 7–15 samples. For each batch, an extraction blank was included using nuclease-free water alongside the samples to serve as negative controls during the extraction process.

To determine whether any potential inhibitors were present in the DNA extracts, we performed a quantitative polymerase chain reaction (qPCR) assay targeting the mitochondrial 16S rRNA gene prior to proceeding with library preparation, thereby ensuring reliable and consistent results throughout the subsequent library preparation process. Since all samples were confirmed to be free of inhibitors, we proceeded with the library preparation using a two-step PCR protocol. For amplification, we used a primer pair targeting the 16S mitochondrial region in mammals (Mam07 primers: MamP007F 5'-CGAGAAGACCCTATGGAGCT-3' and MamP007R 5'-CCGAGGTCTCCCCAACC-3') (Giguet-Covex et al. 2014). We used an equimolar mix of four primers, including three frameshift primers, for both the forward and reverse primers. We used a human-blocking primer in both libraries (Garcés-Pastor et al. 2022; Giguet-Covex et al. 2014) to minimize the amplification of human DNA. We performed eight PCR replicates for each DNA extract, as well as a negative control using nuclease-free water. We ran the PCRs at a final reaction volume of 25 µL as follows: 12.5 µL of AmpliTaq Gold 360 master mix (Applied Biosystems, Waltham, MA, USA), 0.75 µL of each 0.3 µM primer mix, 1 µL of 4 µM human-blocking primer, 2.5 µL of 0.2 µg/µL Bovine Serum Albumin (BSA; Thermo Scientific, Waltham, MA, USA), 1.25 µL of 5% dimethyl sulfoxide (DMSO; Thermo Scientific Pierce), 3.25 µL of molecular grade water, and 3 µL of DNA extract. To check for potential contamination during PCR, we included negative controls alongside the samples. PCR was performed in two separate batches, and each batch included its own negative control. We applied an initial enzymatic activation at 95°C for 10 min when running the PCRs, followed by 40 cycles consisting of denaturation at 95°C for 30 s, annealing at 52°C for 30 s, and elongation at 72°C for 30 s. We conducted the final elongation at 72°C for 5 min.

We pooled all the PCR replicates for each sample and analyzed the pooled samples using agarose gel electrophoresis. We cleaned the samples using AMPure XP beads (Beckman Coulter, Nyon, Switzerland), both before and after indexation PCR. Following the manufacturer's protocol, we performed

indexation PCR using Nextera XT adapters (Illumina, San Diego, CA, USA), with 10 cycles of amplification. Using the fluorescent dsDNA BR Qubit assay (Invitrogen, Waltham, MA, USA) and the Spark M10 plate reader (Tecan, Männedorf, Switzerland), we quantified all samples with both methods before normalizing and pooling them into a single library. We then used capillary electrophoresis to analyze the prepared library, followed by quantification in triplicate through qPCR using the Kapa quantification kit (Kapa Biosystems, Roche, Basel, Switzerland) and the LightCycler 480 Real-Time PCR system (Roche, Basel, Switzerland). Finally, we used the MiSeq Reagent Kit v3 (2 × 75 bp; Illumina) for paired-end sequencing at a theoretical sequencing depth of 200,000 reads per sample, generating an amplicon of approximately 115 bp with the target sequence, primers and adapters.

Following sequencing, the sequences were already demultiplexed. We trimmed the primers using cutadapt, allowing for two mismatches (Martin 2011). Then, we performed quality filtering by removing all reads with more than two expected errors and cutting sequences when the quality dropped below eleven based on the Phred scores of the reads, using the FilterAndTrim function from the R package *dada2* (Callahan et al. 2016). The quality threshold (truncQ = 11) was chosen to ensure high-quality read retention, providing more stringent filtering than the *dada2* default (truncQ = 2). We produced amplicon sequence variants (ASVs) using the *dada2* function implemented in *dada2*, and subsequently removed sequences identified as chimeras using the removeBimeraD-enovo function with the consensus method (also in *dada2*). We then assigned each ASV taxonomically, using the ecotag function from the OBITools toolkit (v1), against a reference database built using CRABs (Jeunen et al. 2023) which combines tailored NCBI queries and the vertebrate EMBL archive (downloaded in June 2024). We validated sequence assignments from ecotag for a given taxonomic level only if the following conditions were met: species (match = 100%), genus (90% ≤ match < 100%), and family (85% ≤ match < 90%; Juhel et al. 2020).

2.5 | Taxonomic Cleaning and Validation

We validated the presence of species detected through eDNA analyses using local checklists (Jamtsho et al. 2021; Koirala and Jamtsho 2019), and we focused exclusively on mammalian species identified through both the eDNA and camera trap methods. We excluded reads corresponding to humans, cattle (*Bos* spp.), goats (*Capra* spp.), horses (*Equus caballus*), and domestic cats (*Felis catus*) detected via eDNA, as well as their occurrences in camera traps, to focus on wild mammals. Due to the lack of comprehensive species-level sequences in the reference database for small mammals, we retained the detections at the genus level, thus considering genus-level presences in the area. We retained species-level identifications when species were listed in a checklist and based on our prior knowledge of its existence in the study area. We constructed a Venn diagram to show the overlap and differences in species detected by eDNA and by camera trapping, providing a comparative assessment of the taxa composition and highlighting the common and unique taxa identified by the two methods.

2.6 | Taxon Accumulation Curves and Detection Probabilities

To investigate how sampling effort influenced the number of taxa identified at all sampling locations, we generated taxon accumulation curves. We fitted accumulation curve using an asymptotic model, implemented with the R package *vegan* v2.8–8 (Oksanen et al. 2024; Veron et al. 2023). We initially fitted curves for all the taxa detected and then constructed curves considering only the taxa that are the main focus of camera trapping in Bhutan, i.e., medium- and large-sized ground-dwelling mammals. We excluded water-dependent taxa, such as *Lutra lutra* and *Latrogele* sp., as well as medium-sized tree-dwelling squirrels as focus species for camera trapping. We categorized medium-sized (2–15 kg) and large-sized (> 15 kg) mammals based on their body weight (Agebo and Tekalign 2022; Francis 2008). We considered all other species weighing less than 2 kg as small mammals for the purposes of this study. We used a Bayesian statistical analysis approach to estimate the detection probability of each species based on 60 L of water sampled for eDNA and 60 days of camera trapping following Lyet et al. (2021). This analysis provided insights into which sampling method demonstrated a higher probability of detecting species when they were present, thus highlighting the relative strengths of eDNA sampling and camera trapping in biodiversity monitoring.

2.7 | Spatial Taxa Composition

To identify and visualize the compositional dissimilarity between the watersheds as detected by the two sampling methods, we computed the β -diversity based on the presence-absence Jaccard dissimilarity index (β_{jac}), using the R package *betapart* v1.6 (Baselga et al. 2023; Baselga and Orme 2012). To disentangle whether the taxonomic similarities between sampling units were driven by taxonomic turnover or by a difference in taxa richness, we decomposed the β_{jac} index into the taxonomic turnover (β_{tu}) and the nestedness (β_{jne}) resultant components (Baselga 2012; Veron et al. 2023). We conducted a principal coordinates analysis (PCoA) on the distance matrix obtained from the Jaccard distance calculation between sites, utilizing the R package *ade4* (v1.7–23) (Dray and Dufour 2007; Thioulouse et al. 2018). Then, we implemented a generalized dissimilarity model (GDM) on the Jaccard dissimilarity matrix to explore the main drivers of taxa compositional differences among sampling sites, using the R package *gdm* (Ferrier et al. 2007). In GDM, predictor variables are transformed using a series of I-spline basis functions and standardized to make the resulting coefficients comparable (Yao et al. 2025). We used dominant climatic and anthropogenic variables shaping the habitats and distributions of species, such as annual mean temperature (°C), annual mean precipitation (mm), distance to the nearest road (m), and settlement density (household density/m²) as predictor variables and species β -diversity as the response variable to construct the GDM. To address potential multicollinearity among the predictor variables, we employed a stepwise procedure, calculating the variance inflation factor (VIF) for each predictor variable at every step of the process, to remove those with a VIF value greater than the predefined threshold of 10 (Naimi 2023) (Table S1). We

fitted the final model by removing the least influential predictor variables, i.e., those with zero coefficients or negligible gradients in the GDM spline curves (Tables S2 and S3). We included geographic distance among sampling sites as one of the predictor variables to assess the influence of pairwise geographic distance on β -diversity.

We conducted all analyses in R (version 4.3.0; R Core Team 2024).

3 | Results

3.1 | Detections From eDNA Metabarcoding and Camera Trapping

Analyses of seven negative controls (five extraction blanks and two PCR blanks) revealed the presence of 11 ASVs, assigned to *Homo sapiens* (human), *Bos taurus* (domesticated cattle), *Rusa* sp., *Sus* sp., *Epicoriculus* sp., and *Sorex* sp. ASVs corresponding to human, domesticated animals, and genus-level assignments of *Rusa* sp. and *Sus* sp. were removed from downstream analyses. However, species-level assignments such as *Rusa unicolor* and *Sus scrofa* were retained. We also retained ASVs assigned to *Epicoriculus* sp. (16, 15, and 6 reads across three extraction blanks) and *Sorex* sp. (one read in one extraction blank) for downstream analyses due to their low reads in controls, as they are also detected in environmental samples. The mean read counts in environmental samples were 2418.18 for *Epicoriculus* sp. and 189.03 for *Sorex* sp. In total, we obtained a total of 5,563,337 sequencing reads, of which 2,816,017 (50.62%) were kept for downstream analyses after bioinformatic filtering. This corresponds to 123 taxonomically assigned ASVs, resulting in 32 species-level and 28 genus-level detections, representing 60 wild mammalian species.

Camera trapping detected 33 species, including small rodents, bats, and two squirrels which could not be identified to the species level but could be classified as distinct species based on morphological differences. Combined, eDNA metabarcoding and camera trapping captured a total of 72 mammalian species (Figure 2A). Among these, eDNA metabarcoding uniquely detected 54.20% ($n = 39$) of the total number of species and camera trapping uniquely detected 16.70% ($n = 12$), while 29.20% ($n = 21$) were detected with both sampling methods. According to IUCN Red List data, 13 of the detected taxa are classified as threatened, with 3 species detected with eDNA metabarcoding, 1 with camera trapping, and 9 with both methods (Figure 2A; Table S4). While two threatened species were only identified to the genus level from eDNA, i.e., *Cervus* and *Lutrogale*, they were most likely the white-lipped deer (*Cervus albirostris*) and the smooth-coated otter (*Lutrogale perspicillata*), as only a single species of each of these genera is known to be present in the area. Overall, a chi-square test of independence showed a marginally non-significant association between the two-detection method and size category ($\chi^2 = 5.91$, $df = 2$, $p = 0.052$). However, standardized residuals indicated that small mammals tended to be detected more frequently by eDNA, whereas large mammals were more frequently detected by camera traps, with no notable difference for medium-sized mammals (Figure S1).

3.2 | Taxon Accumulation Curves and Detection Probability

The asymptotic analyses including all species and those targeted for camera trapping both predicted the highest asymptotic taxa richness (TR_{max}) for the combined sampling methods, followed by eDNA and finally camera trapping (Figure 2B,C). In both analyses, detection from eDNA showed a higher initial taxa richness (18.96 and 9.56, respectively) and a higher rate of accumulation (1.5 and 0.67, respectively; Table S5). When all taxa were considered, a single eDNA site (with pooled site replicates) captured the same taxa richness as an average of 5.79 camera-trap stations. Similarly, when taxa assigned at the species level and targeted for camera trapping were considered, a single eDNA site captured the same taxa richness as approximately 2.43 camera-trap stations on average. The average detection probability of taxa identified through eDNA ranged from 0.05 to 0.94, with a mean of 0.27 ± 0.06 , and the detection probability from camera trapping ranged from 0.008 to 0.03 (mean = 0.10 ± 0.02 ; Figure 3). The highest detection probabilities from eDNA metabarcoding were observed for wild boar (*Sus scrofa*; 0.94 ± 0.05), sambar deer (*Rusa unicolor*; 0.88 ± 0.07), and Asiatic black bear (*Ursus thibetanus*; 0.85 ± 0.07). For camera trapping, the species with the highest detection probabilities were the Asiatic golden cat (*Catopuma temminckii*; 0.21 ± 0.03), followed by barking deer (*Muntiacus muntjak*; 0.21 ± 0.03), wild boar (0.20 ± 0.03), Asiatic black bear (0.19 ± 0.03), and sambar deer (0.19 ± 0.03). For all taxa, whether grouped by order or size category, the mean detection probabilities were higher for eDNA metabarcoding than for camera trapping (Figure S2A,B).

3.3 | Spatial Taxa Composition

Camera trapping recovered 1–15 taxa per station, while eDNA metabarcoding detected 7–26 taxa per site. The average site-level taxa richness was higher for eDNA (15.40 ± 5.84) than for camera traps (5.78 ± 3.61). The average dissimilarity in taxa composition between the sampling sites, measured using the Jaccard dissimilarity index (β_{jac}), was high for both methods but differed in magnitude and underlying components. For camera trapping, β_{jac} was 0.85 ± 0.19 , reflecting high dissimilarity driven mainly by taxa turnover ($\beta_{jtu} = 0.73 \pm 0.32$) rather than nestedness ($\beta_{jne} = 0.12 \pm 0.22$). eDNA metabarcoding showed a lower β_{jac} (0.59 ± 0.17) compared with camera trapping, with the value again mainly driven by taxa turnover ($\beta_{jtu} = 0.43 \pm 0.20$) rather than nestedness ($\beta_{jne} = 0.17 \pm 0.14$). Further, the ordination results conducted on the dissimilarity matrix indicated clear spatial structuring, explaining 43.24% of the total inertia in the eDNA data (Figure 4A) and 31.00% in the camera trap data (Figure 4B). The highland watersheds (Laya and Lingzhi) exhibited distinct taxa composition compared with lowland watersheds (Thimphu, Goenshari, Phochu and Dangchu) for both sampling methods (Figure 4C,D). This corresponded to the GDM I-spline curve, where low temperatures were associated with highland habitats and with greater distances to roads (Figure S3A,B).

GDMs conducted to explore the effects of the geographic distance among sampling sites and of the selected environmental gradients on β -diversity explained 39.65% of the deviance in an

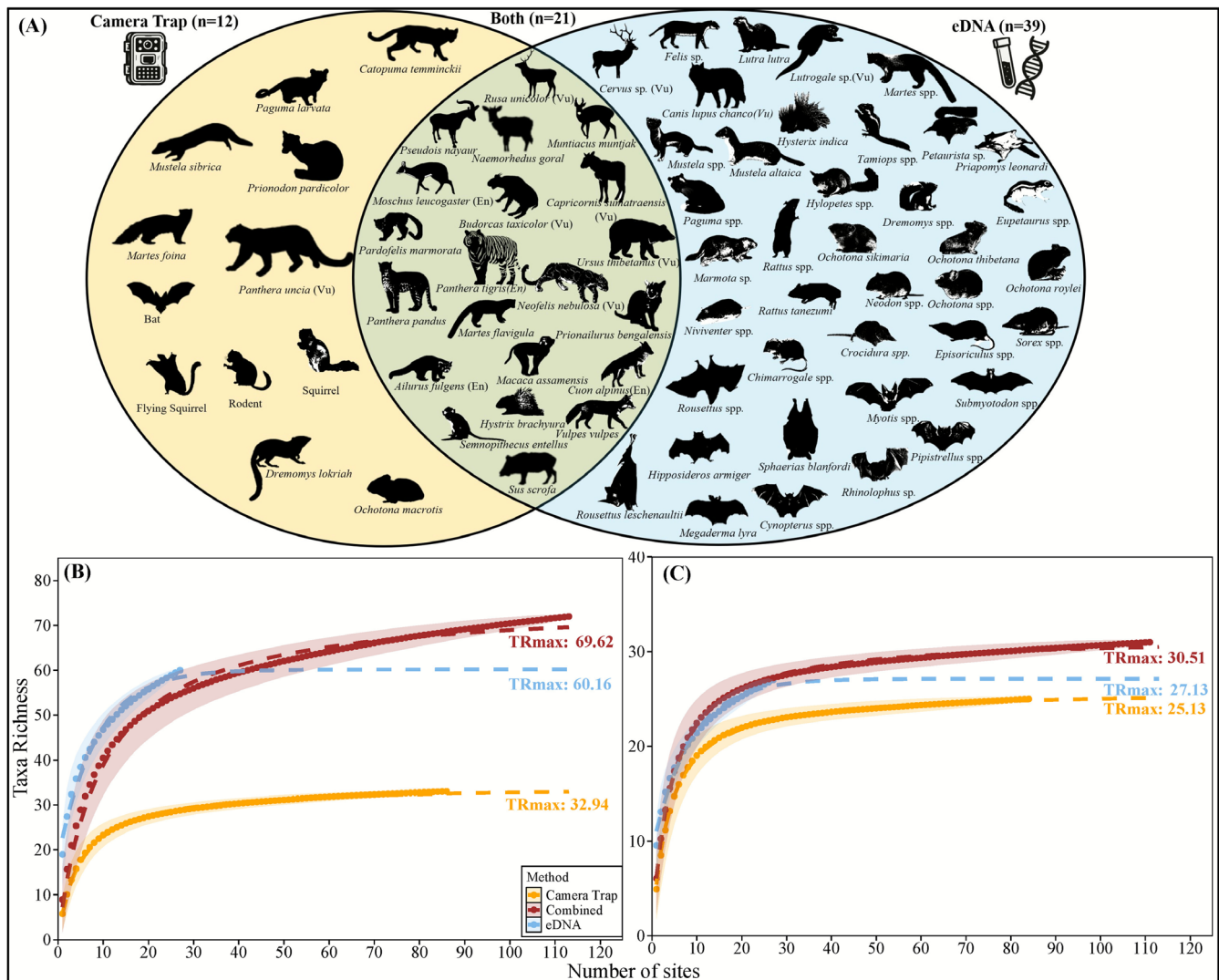


FIGURE 2 | Representation of taxa detection using a Venn diagram and taxon accumulation curves. (A) Venn diagram illustrating the number of taxa detected using eDNA metabarcoding, camera trapping, and both methods combined. IUCN Red List threatened species categories are indicated in brackets: Vulnerable (Vu), endangered (En), and critically endangered (CR). (B) Taxon accumulation curves for all species detected with eDNA metabarcoding, camera trapping, and both sampling methods, showing the asymptotic richness (TR_{max}). (C) Taxon accumulation curves for medium- and large-sized ground-dwelling mammals, which are primarily targeted in camera-trap surveys, excluding water-dependent species.

eDNA model and 31.21% of the deviance in a camera trap model (Figure S3A,B). In both GDMs, annual mean air temperature (°C) predicted the highest taxa dissimilarity between sites, followed by distance to the nearest road (m). In the eDNA-based GDM, geographic distance among sampling sites was the third most important predictor. In the camera trap GDM, this geographic distance contributed minimally to model deviance, with the third most important predictor being annual mean precipitation (mm).

4 | Discussion

The application of eDNA metabarcoding across the complex terrain of north-western Bhutan provides a comprehensive overview of mammalian composition, comparable to that obtained with camera trapping. We found that eDNA surveys detected a large number of mammalian species, including many species not recorded by camera traps. Moreover, the detection

probabilities of most species were higher with eDNA than with camera traps. Further, the similar spatial ordination of taxa composition between lowland and highland communities for eDNA metabarcoding and camera trapping highlights the importance of considering watershed-based eDNA sampling to provide a meaningful interpretation of the spatial distribution of species. Overall, our analyses support the use of eDNA metabarcoding as an approach for rapidly surveying mammals, particularly in Bhutan, where inaccessible terrain poses challenges for camera trapping and extensive river networks offer unique opportunities for monitoring based on eDNA in water samples.

Detection methods are generally associated with different performances for distinct functional groups (Leempoel et al. 2020). eDNA detected many medium- and large-sized mammalian species also recorded by camera trapping, and both methods performed similarly, detecting most of the IUCN threatened species. However, eDNA performed significantly better for small mammal orders such as Chiroptera, Eulipotyphla and

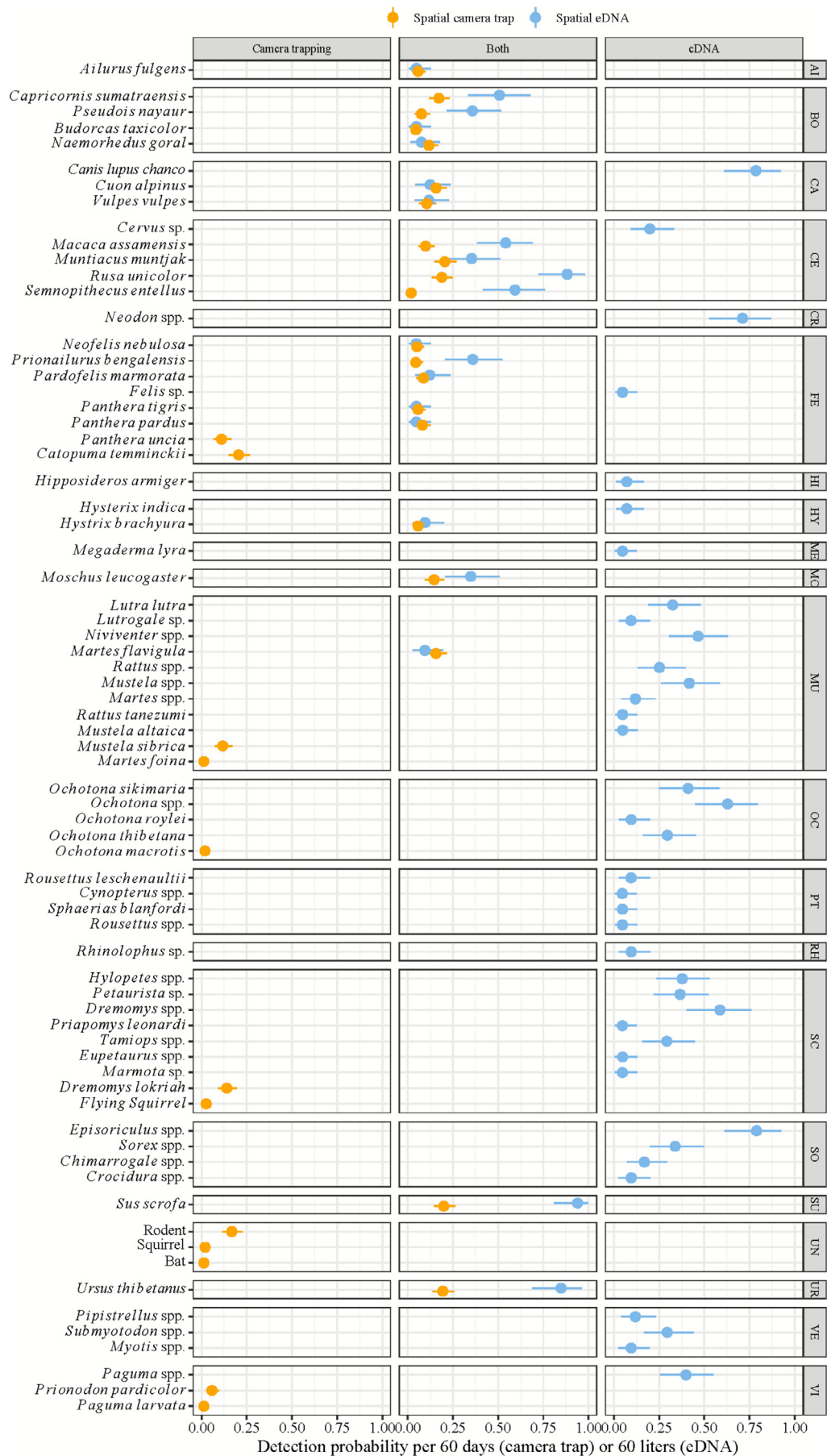


FIGURE 3 | Legend on next page.

FIGURE 3 | Overall species detection probabilities. The circles indicate the average detection probabilities for the species, while the horizontal bars represent the limits of the 95% Bayesian credibility interval. We applied the same approach as in Lyet et al. (2021) to calculate detection probabilities. Taxa are grouped by family (shown in the far-right panel), with family names abbreviated. Full family names are provided in Table S6.

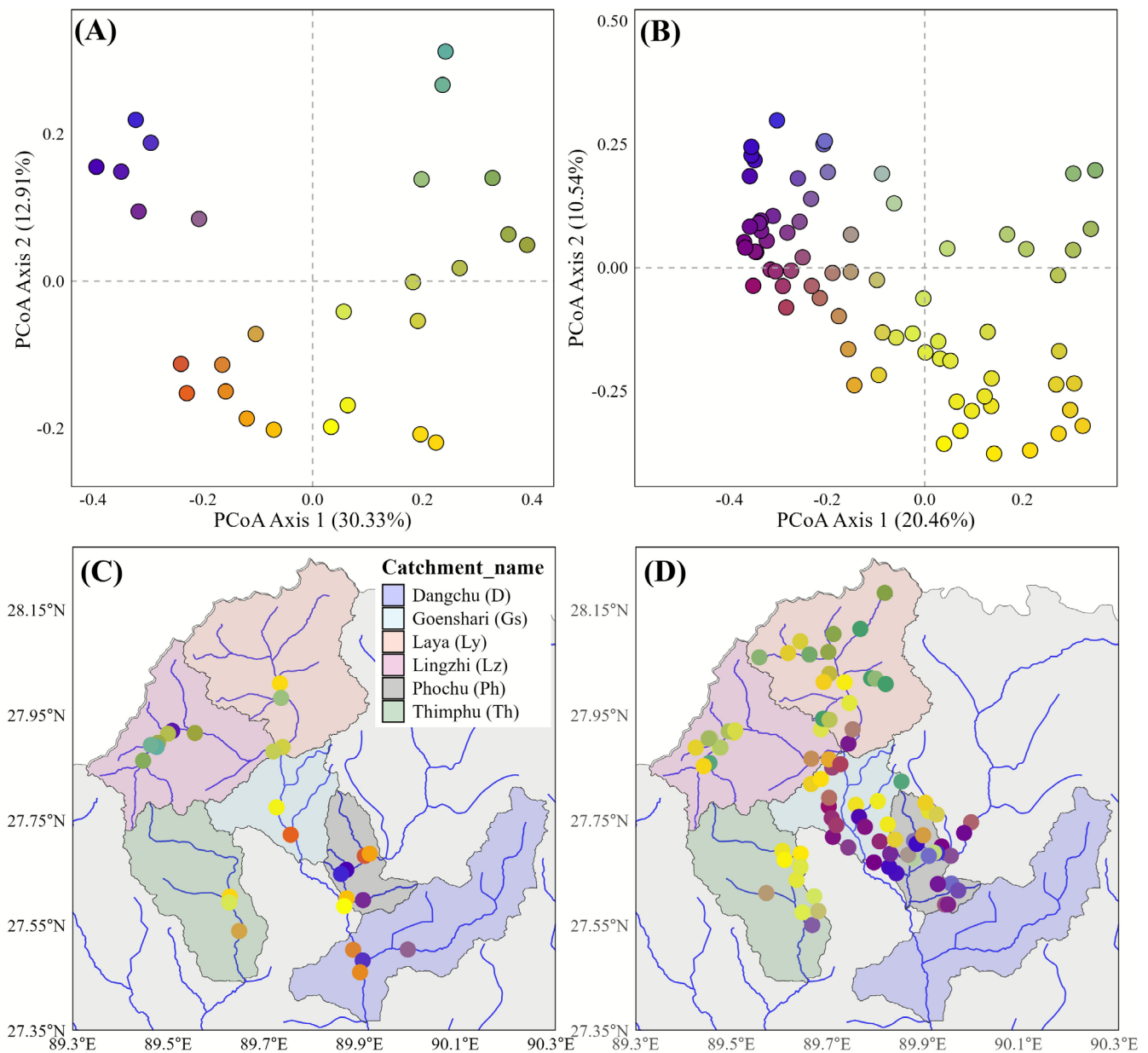


FIGURE 4 | Study area and principal coordinates analysis (PCoA) of taxa composition. (A) Ordination plot of taxa composition dissimilarity among eDNA sampling sites in different watersheds. (B) Ordination plot of taxa composition dissimilarity among camera trap stations in different watersheds. Dissimilarity matrices were constructed using the Jaccard index based on taxa presence (1) and absence (0) data at each site. (C) Ordination map of eDNA-based taxa composition across the watersheds. (D) Ordination map of camera-trap-based taxa composition across the watersheds. Colors represent the positions in the 2D PCoA space.

Lagamorpha (Figure S2A). This difference in detection ability is probably due to the fact that camera traps are set up at a height of 30–45 cm above the ground (DoFPS 2023; NCD 2023); this set-up has been found to be effective in detecting medium- and large-sized mammals, but small mammals are less likely to trigger the camera traps (Mas-Carrió et al. 2022). Moreover, water-dependent species of small mammals are also less likely to trigger the camera traps, as the cameras are placed further

away from water bodies. Further, the detection probabilities of species from eDNA were higher than those from camera traps. Our findings agree with those from previous studies comparing the two methods (Di Girolamo et al. 2024; Leempoel et al. 2020; Mas-Carrió et al. 2022; Sales et al. 2020) and suggest that camera traps alone are insufficient for monitoring mammal diversity. While the likelihood of detection with the different methods is influenced by various factors (e.g., presence of DNA in water for

eDNA or movement of animals in front of cameras for camera traps), eDNA demonstrates a strong potential of detecting mammalian species of all sizes, even if some species were only detected by camera traps.

Although both methods have strengths and limitations, we also found that eDNA and camera trapping complement each other in species detections. At several locations, species not detected by one method were successfully detected by the other method. For example, despite broader coverage of eDNA sampling in the Lingzhi watershed, the habitat of the snow leopard (*Panthera uncia*) (Dendup and Lham 2024; Lham et al. 2021), the species was only detected through camera traps. Conversely, while the royal Bengal tiger (*Panthera tigris tigris*) primarily inhabits lowland areas (DoFPS 2023), eDNA failed to detect the species in these areas but did detect it in highland areas (> 3000 m), where it has rarely been reported (Dendup and Lham 2024). In particular, large-bodied felids occur at lower densities at high elevations (Lyet et al. 2021) and often spend time on the ridges. This could lead to them being less likely to be detected with eDNA obtained from filtered water. However, longer sampling times and more replicates can increase the likelihood of detections (Deiner and Altermatt 2014; Lyet et al. 2021; Merkes et al. 2014). The higher proportion of mammals detected from eDNA than from camera traps in our current study suggests the great potential of using eDNA in Bhutan. With the presence of extensive networks of rivers in the country, watersheds present a well-defined ecological sampling unit for eDNA-based studies. By using eDNA to screen for species presences at the watershed level, Bhutan could develop an efficient, cost-effective and scientifically robust wildlife monitoring programme.

Most monitoring methods, such as camera trapping, correspond to point measurements, but eDNA metabarcoding allows a larger fraction of the landscape to be screened (Chacko et al. 2023; Ushio et al. 2017). Our results demonstrate that eDNA sampling based on watersheds can provide accurate information on the spatial distribution and habitat use of the species, with the detections based on eDNA showing similar relationships with environmental variables compared with the detections from camera traps. Specifically, the spatial analyses of taxa composition for both eDNA metabarcoding and camera trapping indicated similar distributions between highland and lowland habitats, mainly separated by temperature, corresponding to faunal turnover along the elevation gradient (Snider et al. 2024). The distinction in taxa composition is well marked by the assigned watersheds, where the Lingzhi and Laya watersheds are alpine habitats and the others are lowland habitats characterized by broadleaved forests. Taxa composition was further associated with distances to roads, which correspond to a gradient of anthropogenic disturbances that can impact wild mammal occupancy (Penjor et al. 2021), suggesting that we can use eDNA to quantify the human impact on wildlife. Chacko et al. (2023) reported a positive correlation of vertebrate taxa composition from eDNA with catchment-level forest and grassland cover, emphasizing how the habitat types shape taxa diversity and composition. These findings highlight the importance of watershed-based eDNA sampling, where detections correspond to species presences within each watershed or habitat. This approach addresses the challenges of large-scale monitoring, where different habitats complicate species detections and habitat associations (Urycki

et al. 2024). However, the downside is that the more diffuse eDNA signal over larger areas might lead to weaker relationships with environmental predictors in spatial models (Carraro et al. 2020).

Despite our efforts to compare the detection capabilities between eDNA and camera trapping, the study has certain limitations. While we collected eDNA samples from the main river of each watershed, only a few river tributaries were selected, which may have resulted in underrepresentation of certain areas. Expanding sampling efforts to include more interconnected streams could provide a more comprehensive representation of the taxa composition in each watershed. Further, filtering the water for a longer period can improve the recovery of eDNA residues. In our sampling effort, we filtered water for 1 to 2 h when clogging was not an issue. However, during the monsoon, some filters clogged quickly due to high sediment loads, limiting the filtration time. In contrast, during winter, when the water was clearer, filtration could proceed for the full 1 to 2 h duration in most rivers or streams, but mammals might have been less active during this season (Guiden and Orrock 2020), thus shedding less of their DNA into the water. We recommend increasing the number of replicates during the monsoon season, when filters are more prone to clogging, and enhancing the overall sampling effort across all seasons. Recent work highlights that both larger volumes of filtered water and longer durations of filtration increase the probability of detection (Lyet et al. 2021). Additionally, Bhutan has extensive records of medium- and large-sized mammals from previous camera trapping studies, and their DNA reference sequences are well represented in existing databases. However, research on small mammals remains limited. Except for a few common species, most small mammals could only be identified to the genus level in the eDNA results, indicating gaps and incomplete representation in the current reference databases. Since eDNA frequently detects small mammals, developing a complete and reliable checklist and complementing reference databases for these species would enhance the accuracy of future eDNA studies.

5 | Conclusion

Here, we have demonstrated the capabilities of watershed-based eDNA sampling from rivers and streams in comparison to camera trapping to detect mammalian species in north-western Bhutan. Our study showed that eDNA surveys detected many mammalian species that were also detected by camera traps, including threatened species. We also found that detection probabilities of most species from eDNA metabarcoding exceed those from camera trapping, showing the strength of eDNA in detecting the species more efficiently. Further, the spatial analyses of taxa composition for eDNA and camera trapping showed similar taxa compositions between highland and lowland habitats, highlighting the importance of considering watershed-based eDNA sampling to provide a meaningful interpretation of the spatial distribution of species. Although our study pointed to the strengths and limitations of both methods in detecting the species, it suggests that eDNA—when used with increased sampling efforts and larger volumes of water—could emerge as a robust tool for biodiversity assessments. This can be particularly advantageous in mountainous countries like Bhutan, which

are characterized by rugged terrain, and where most places are remote and inaccessible. Due to this, most areas are underrepresented in the camera trapping setup. In such cases, eDNA can become an important tool to detect species across large catchments, and, as our study emphasized, watershed-based sampling can further enhance the resolution of environmental signals and clarify their origins. As a global biodiversity hotspot with ecologically rugged terrain intertwined with extensive hydrological networks, Bhutan is in a unique position to utilize the advantages of eDNA for biodiversity monitoring. This approach could not only enhance our understanding of ecological systems but also cost-effectively inform practitioners on how to safeguard and manage biodiversity.

Author Contributions

K.S., L.P. and C.A. designed the study, performed statistical analyses, and wrote the first draft of the manuscript. S.T., M.L. and V.M. were involved in eDNA analysis and bioinformatics analyses. A.L. provided support for statistical analysis. T.D. and L.D. were involved in the management and cleaning of past camera trap data. All authors discussed the results and contributed to the development of the manuscript.

Acknowledgments

This work was supported by the Sawiris Foundation for Social Development, ETH4D, and the Ecosystems and Landscape Evolution (ELE) group of ETH Zürich, Switzerland. Moreover, we thank the Department of Forest and Park Services (DoFPS) of Bhutan for granting us permission to carry out this work and use past camera trap data to address our research questions. We are grateful to everyone who helped us during this work. We also thank Ms. Melissa Dawes for her proofreading and improvements to the English text of this manuscript. Open access publishing facilitated by Eidgenössische Technische Hochschule Zurich, as part of the Wiley - Eidgenössische Technische Hochschule Zurich agreement via the Consortium Of Swiss Academic Libraries.

Declaration of generative AI and AI-assisted technologies in the writing process: During the preparation of this work, the authors used ChatGPT and REF-N-WRITE to improve the readability and language of the manuscript. After using these tools, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

Funding

This work was supported by ETH4D.

Ethics Statement

Permission to conduct this study was obtained from Department of Forest and Park Services (DoFPS), Ministry of Energy & Natural Resources, Bhutan. The study was approved by the Research Steering Committee on 14 April 2023 under Application ID: 330104. Prior to conducting the survey, the objectives of the research were presented at the Divisional Forest Office, Punakha and Jigme Dorji National Park (JDNP), Bhutan.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Envidat at <https://www.envidat.ch>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Assessment of multicollinearity using variance inflation factors (VIFs) for environmental and anthropogenic predictor variables in generalized dissimilarity models (GDMs) based on eDNA and camera-trap data. **Table S2:** Variables used in a final generalized dissimilarity model (GDM) for eDNA-based species detection. Variables are ranked by importance according to their summed model coefficients. The final model was fitted by removing the least influential predictors, i.e., those with zero coefficients or negligible gradients in the GDM spline curves. **Table S3:** Variables used in a final generalized dissimilarity model (GDM) for camera-trap-based species detection. Variables are ranked by importance according to their summed model coefficients. **Table S4:** List of IUCN threatened species detected by eDNA and camera trapping. Species detected by both methods are indicated as "Both". **Table S5:** Results of asymptotic analyses for all the species detected with eDNA metabarcoding, with camera trapping, and with the two methods combined, as well for those species usually targeted for camera trapping, i.e., ground-dwelling terrestrial medium- and large-sized mammals. **Table S6:** List of species with their corresponding taxonomic families with family abbreviations. **Figure S1:** Detection frequencies of species based on body size categories. The frequency refers to the occurrence of species in one sampling site as one occurrence. **Figure S2:** Comparison of mean detection probabilities of species grouped by (A) order and (B) size category. Error bars represent the standard deviation. **Figure S3:** Results from the generalized dissimilarity model (GDM). The main plots show the relationship between observed compositional dissimilarity and predicted ecological distance from (A) eDNA and (B) camera trapping. The smaller panels on the right show the GDM I-spline partial functions for the top three most influential predictor variables based on the summed model coefficient, depicting the rate of species turnover along environmental gradients and geographic distance among sampling sites, retained in the model.